Rapid Rotation and Cubic Damping in Incompressible Navier-Stokes Flows

A Carefully Scoped Galerkin Averaging Framework on the Periodic Three-Torus

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Research note / mathematical framework | 31 October 2026

Abstract. We formulate the rapidly rotating, cubically damped incompressible Navier-Stokes system on the periodic three-torus and distinguish its exact filtered equation from its formal resonant average. The Coriolis operator is diagonalized on each solenoidal Fourier mode, including both inertial-wave branches and the zero spatial mode. We identify the kernel as a 2D-3C slow subspace, and give the correct branch-indexed resonance conditions for quadratic advection and cubic damping. We prove an energy inequality for finite Fourier-Galerkin approximations, uniform in the Rossby parameter. We then state and prove a finite-dimensional averaging theorem: for each fixed truncation and smooth forcing, the filtered Galerkin solution converges on finite time intervals to the averaged Galerkin ODE, with an O(epsilon) error whose constant may depend on the truncation. This result is deliberately not presented as a convergence theorem for the full PDE. Uniform estimates in the truncation parameter, a PDE-level singular-limit theorem, and a physical constitutive derivation of the cubic coefficient remain open tasks.

Keywords: rotating fluids; Navier-Stokes; cubic damping; inertial waves; resonant averaging; Fourier-Galerkin approximation

Scope statement. This note establishes a finite-dimensional theorem and records a program for the infinite-dimensional problem. It does not claim a new global-regularity theorem for three-dimensional Navier-Stokes, nor a PDE-level fast-rotation limit.

1. Introduction and scope

Rapid rotation introduces inertial-wave oscillations whose phases interact through the nonlinear terms. A rigorous limiting description must retain resonant interactions; replacing the limit by a two-dimensional equation without analyzing the spectrum is not justified on a general periodic three-torus. Separately, the cubic damping term has an established role in global regularity theory for damped three-dimensional Navier-Stokes equations. The question pursued here is narrower: how to define and analyze the filtered resonant dynamics for the combined rotating and cubically damped model.

The contribution of this note is intentionally limited. First, it gives a consistent spectral bookkeeping for the projected Coriolis operator, including the mean mode. Second, it proves the exact finite-dimensional averaging statement for Fourier-Galerkin systems. The result is a rigorous baseline for later estimates uniform in the Fourier cutoff. Such uniformity is essential before passing from Galerkin ODEs to the PDE and is not proved here.

Global strong solvability for the unrotated damped system at cubic exponent is already known; the damping component is therefore not claimed as novel. See Zhou (2012) for the cubic threshold result and Cai and Jiu (2008) for earlier weak and strong solution theory under their stated exponent assumptions.

2. Model, operators, and exact energy balance

Let T^3=(R/2pi Z)^3, and let u^epsilon be a real, divergence-free vector field. We use the Leray projection P onto divergence-free fields and write the projected equation as

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d_t u^epsilon + nu A u^epsilon + B(u^epsilon,u^epsilon) + (1/epsilon)L u^epsilon + gamma
                                  C(u^epsilon) = f,
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where A=-P Delta, B(a,b)=P[(a dot grad)b], L=P(e_3 cross .), and C(a)=P(|a|^2 a). Here nu>0, gamma>0, and epsilon is the Rossby number. The spatially constant mode is retained unless explicitly stated otherwise. All inner products below are over T^3.

For smooth solutions, taking the L2 inner product with u^epsilon gives the exact balance

```
(1/2) d_t ||u^epsilon||_2^2 + nu ||grad u^epsilon||_2^2 + gamma ||u^epsilon||_4^4 =
                                    (f,u^epsilon).
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The transport term vanishes by incompressibility and periodicity. The Coriolis term vanishes because L is skew-adjoint on the solenoidal L2 space. The damping contribution is (C(u),u)=integral |u|^4. In a Fourier-Galerkin approximation these cancellations remain exact when the projection is orthogonal and the nonlinear terms are projected back into the same solenoidal space.

For f in L^2(0,T;L^2), finite volume and Young inequality give, with a constant depending on gamma and the torus,

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 sup_[0,T] ||u^epsilon||_2^2 + nu integral_0^T ||grad u^epsilon||_2^2 dt + gamma
integral_0^T ||u^epsilon||_4^4 dt <= C(T,nu,gamma,||u_0||_2,||f||_{L^2_t L^2_x}),
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with C independent of epsilon. This is an energy estimate, not an H1 supremum estimate. No uniform-in-epsilon H1 estimate for the full PDE is asserted in this note.

3. Correct spectral decomposition of the Coriolis operator

For a nonzero Fourier wavevector k in Z^3, the solenoidal Fourier fiber is k-perp and the Leray symbol is P_k=I-k tensor k/|k|^2. The restriction of P_k(e_3 cross .) to this two-dimensional plane is skew-symmetric and has eigenvalues

```
lambda_(k,sigma) = i omega_(k,sigma), omega_(k,sigma) = sigma k_3/|k|, sigma in {+1,-1}.
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Thus there are two inertial-wave branches. When k_3=0, both eigenvalues are zero: the entire solenoidal fiber at that wavevector belongs to the kernel. At the spatial mean k=0, the Leray projection is the identity and e_3 cross . has eigenvalues +i, -i on horizontal mean velocities and 0 on the vertical mean velocity. This mode must be treated separately; the formula k_3/|k| does not apply at k=0.

Consequently, ker L consists of all solenoidal modes with k_3=0 together with the vertical spatial mean. It is a 2D-3C slow subspace: fields are independent of x_3, but may have three velocity components. Horizontal mean velocity is not slow; it rotates at unit frequency under L.

Choose a normalized helical eigenbasis h_(k,sigma) for k nonzero, with the above eigenvalues. For interactions among nonzero modes, the quadratic resonance condition for input modes (m,sigma_m),(n,sigma_n) and output mode (k,sigma_k) is

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k=m+n, omega_(k,sigma_k) = omega_(m,sigma_m) + omega_(n,sigma_n).
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The cubic damping has a corresponding four-mode condition

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k=l+m+n, omega_(k,sigma_k) = omega_(l,sigma_l)+omega_(m,sigma_m)+omega_(n,sigma_n).
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Zero-frequency modes and the k=0 branches are included by assigning their actual eigenfrequencies, rather than by substituting into k_3/|k|. These phase conditions identify candidate resonances; the associated vector interaction coefficient can still vanish and must be evaluated separately.

4. Filtering and the averaged Galerkin system

Let P_N denote the orthogonal projection onto solenoidal Fourier modes with |k|<=N, including the spatial mean. Set A_N=P_N A, L_N=P_N L, B_N=P_N B, and C_N=P_N C, with all arguments restricted to the finite-dimensional space H_N. The Galerkin system is

```
d_t u_N^epsilon + nu A_N u_N^epsilon + B_N(u_N^epsilon,u_N^epsilon) + (1/epsilon)L_N
                    u_N^epsilon + gamma C_N(u_N^epsilon) = f_N(t).
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Since A_N commutes with L_N, define the filtered variable v_N^epsilon(t)=exp(t L_N/epsilon)u_N^epsilon(t). Its exact equation is

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d_t v_N^epsilon = F_N(t/epsilon,t,v_N^epsilon),
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where, writing tau=t/epsilon,

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F_N(tau,t,v) = -nu A_N v - exp(tau L_N) B_N(exp(-tau L_N)v,exp(-tau L_N)v)

- gamma exp(tau L_N) C_N(exp(-tau L_N)v) + exp(tau L_N) f_N(t).
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The averaged vector field is the Cesaro mean over the compact group orbit

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Fbar_N(t,v) = lim_(R to infinity) (1/R) integral_0^R F_N(s,t,v) ds.
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In finite dimension the orbit is a finite-frequency quasiperiodic function; its Cesaro mean exists and consists exactly of the zero temporal-frequency terms. In particular, the averaged forcing is exp(t L_N) averaged over the group, which equals P_(ker L_N) f_N(t), not f_N(t) in general. The averaged Galerkin equation is

```
d_t v_N = Fbar_N(t,v_N), v_N(0)=P_N u_0.
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5. A finite-dimensional averaging theorem

Theorem 5.1 (fixed-cutoff filtered averaging). Fix N0. Let P_N u_0 be divergence-free and let f_N belong to C^1([0,T];H_N). Let v_N^epsilon solve the filtered Galerkin equation above, and let v_N solve the averaged equation with the same initial value. Then both solutions exist on [0,T], and there is a constant C_(N,T), independent of epsilon in (0,1], such that

```
sup_(0<=t<=T) ||v_N^epsilon(t)-v_N(t)||_2 <= C_(N,T) epsilon.
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The constant may depend on N, T, nu, gamma, u_0, and the C^1 norm of f_N. In particular, this theorem gives no bound uniform as N tends to infinity.

Proof. For fixed N, H_N is finite-dimensional. The filtered vector field is a polynomial in v with coefficients that are finite sums of exponentials exp(i lambda tau); the forcing has the same form with slowly varying C^1 amplitudes. Let G_N(tau,t,v) be the zero-mean primitive in tau of F_N(tau,t,v)-Fbar_N(t,v). Since only finitely many frequencies occur at this fixed cutoff, every nonzero frequency has a positive minimum absolute value. Dividing each nonzero Fourier coefficient by its frequency defines G_N, which is bounded together with its first derivatives on bounded subsets of [0,T] x H_N.

The energy identity for u_N^epsilon gives a bound on ||u_N^epsilon||_2 over [0,T] independent of epsilon. Since exp(t L_N/epsilon) is unitary, the same bound holds for v_N^epsilon. The averaged equation has the corresponding averaged energy inequality, so v_N also remains bounded on [0,T]. On a common bounded set the vector fields and their derivatives are bounded and locally Lipschitz.

Use the near-identity variable w_N^epsilon=v_N^epsilon-epsilon G_N(t/epsilon,t,v_N^epsilon). Differentiation cancels the zero-mean oscillatory part and gives d_t w_N^epsilon=Fbar_N(t,v_N^epsilon)+O_(N,T)(epsilon), uniformly on [0,T]. Also ||w_N^epsilon(0)-P_Nu_0||_2=O_(N,T)(epsilon). Local Lipschitz continuity of Fbar_N and Gronwall inequality yield ||w_N^epsilon-v_N||_(L-infinity_t H_N)=O_(N,T)(epsilon). Since ||v_N^epsilon-w_N^epsilon||=O_(N,T)(epsilon), the stated estimate follows. This is the standard finite-dimensional averaging argument; the fixed N dependence is essential. End of proof.

Interpretation. The theorem rigorously validates the filtered and resonant construction for each fixed Fourier truncation. It is not a new PDE singular-limit theorem and does not imply convergence of the unfiltered velocity. A PDE limit would require estimates and averaging remainders controlled uniformly in N, including the small-divisor structure of the lattice.

6. Physical dimensions and status of the damping coefficient

In dimensional variables, each acceleration term has dimensions L T^-2. Since |u|^2u has dimensions L^3 T^-3, the coefficient in gamma |u|^2u must satisfy

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[gamma] = T L^-2 (for example, s m^-2).
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Dimensional consistency does not identify a physical mechanism. The cubic term is treated here as a phenomenological nonlinear dissipation. It should not be called a standard Forchheimer law without qualification: classical Darcy-Forchheimer closures commonly contain a quadratic drag proportional to |u|u. A cubic velocity law requires its own constitutive derivation, material setting, or explicit empirical justification.

A defensible empirical program would (i) derive the candidate drag from a specified pore-scale or volume-averaged model, (ii) fit gamma using only a calibration set, (iii) evaluate held-out predictions with measurement uncertainties, and (iv) compare against undamped Navier-Stokes and appropriate linear/quadratic drag baselines. Numerical agreement alone calibrates a closure; it does not establish its microscopic cause.

7. Research program and limitations

The next mathematical task is to seek estimates for the Galerkin averaging remainder whose constants remain controlled as N increases. The proof must specify a solution class, regularity of the initial data and forcing, the domain geometry, the treatment of zero modes, and the precise norm and time interval of convergence. Resonant coefficients should be computed in the helical basis, rather than inferred from phase matching alone. Any quantitative rate must state its dependence on N and on the regularity norms.

The current theorem is deliberately finite-dimensional. It establishes neither global H1 estimates for the full rotating PDE nor a strong PDE-level limit as epsilon tends to zero. Existing fast-rotation analyses show that resonances, domain geometry, and small divisors matter; the averaged dynamics cannot be presumed to be a simple two-dimensional Navier-Stokes equation.

The present note is therefore a scoped mathematical framework with one standard but fully stated finite-dimensional theorem. A claim of novelty should be made only after a systematic comparison with the existing fast-rotation averaging literature. The prospective contribution is the coupled resonant effect of cubic damping, if a PDE-level theorem or a genuinely new uniform estimate can be proved.

References

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Publication status. This document is a research framework, not a submission-ready claim of a new PDE theorem. The finite-cutoff theorem is included with its proof and explicit scope; the infinite-dimensional limit remains an open objective of the program.